A Constructive Introduction to Quantum Field Theory

Lecture 7: Φ^4_3 on the torus III. UV stability, compactness and the limiting FBSDE

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November 2026 — provisional

Abstract

We close the argument. At a fixed lattice the error of the approximate forecast vanishes at infinite resolution, because the explicit action matches the bare action there; this removes the boundary term of the energy estimate, and gives bounds uniform in the lattice (UV stability). The order of limits is essential: first the upper scale at fixed lattice, then the lattice. The uniform energy bounds give compactness of the forecast in space and scale, by a maximum principle and a linear elliptic estimate, and tightness of the lattice measures in H^{-s}, $s > \frac{1}{2}$. Identifying the limit requires two further arguments: weak convergence of the deterministic coefficients against compact resolvents, and strong compactness of the backward integrand, which follows from the positivity of the posterior covariance and its exact Riccati equation. The result is a solution of the FBSDE on the continuum torus, driven by a Brownian motion obtained as the limit of the lattice innovations, with no additional noise, whose terminal law is a subsequential limit of the lattice measures and satisfies a renormalised integration by parts formula. We end with what is not proved and with the infinite-volume problem.

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Introduction

Lecture 6 proved, at every lattice cutoff N and every upper scale T,

$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \le C_\gamma\,(1 + p^5 + \delta_{N,T,p}), \qquad \delta_{N,T,p} = \|T\,Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}, \tag{1}$$

for the backward energy E_T and the large-field action J_T of (15.6). Three things remain.

i. Show that the boundary term vanishes: $T\,Z_T \to 0$ as $T \to \infty$ at fixed N (Section 18).

ii. Turn the uniform energy bounds into compactness of the processes and of the measures (Section 19).

iii. Identify the limit: the deterministic coefficients, the noise, and the equations (Section 20).

The structure of (iii) is instructive: convergence of the fields alone does not suffice to pass to the limit in an FBSDE, because the backward martingale might converge to something with *extra noise*. Here the statistical meaning of the backward variable, a posterior mean with a positive covariance, saves us.

18 The boundary at infinite resolution

18.1 The boundary lemma

At a fixed lattice, the terminal resolution is $q = \infty$, and the field at resolution q, $\Phi_q = C_q\,P_q$, converges to the terminal field Φ_∞. The approximate action $\bar\Gamma_q$ was built to converge to the bare action: $\mu_q \to \mu_N$, $\Theta_q \to 0$, and the nonlinear corrections were integrated from $q = \infty$, so that they vanish there. This is the boundary condition of the backward equation.

Lemma 18.1. (Boundary lemma) *At fixed N, for every $p < \infty$, as $q \to \infty$,*

$$\|Y_q - \Phi_\infty\|_{L^p(\mathbb{P};H_N)} + \|q\,Z_q\|_{L^p(\mathbb{P};H_N)} \le C_{N,p}\,q^{-1/2}, \qquad \textit{hence} \qquad \delta_{N,T,p} \le C_{N,p}\,T^{-1}. \tag{18.1}$$

The constants depend on N; this is not a uniform estimate.

Proof. All norms on H_N are equivalent, and we do not track the dependence on N.

The field at resolution q. By (12.5), $\Phi_q = C_q P_q = (q - q_0)\,C_q\,\Phi_\infty + C_q\,B_{q-q_0}$, and $(q-q_0)\,C_q - I = -C_q\,K_{q_0} = O(q^{-1})$, $\|C_q\,B_{q-q_0}\|_{L^p} = O(q^{-1/2})$. So $\|\Phi_q - \Phi_\infty\|_{L^p} \le C_{N,p}\,q^{-1/2}$.

The approximate forecast. Global convexity gives $\|Y_q\| \le C\|P_q\|/q$, bounded in every L^p uniformly in q. The coordinate equation $\Phi_q = Y_q + C_q\,D\bar\Gamma_q(Y_q)$ and the bounds $|\mu_q| \le C_N$, $\|\Theta_q\|_{\mathrm{row}} = O_N(q^{-1/2})$ and $\|D\tilde h_q(y)\|_\infty \le C\mathsf{S}_4(h_q^{[2]})\|y\|^3_\infty \le C\gamma^2 q^{-1/2}\|y\|^3_\infty$ (Taylor from the fourth order, since the low jets vanish) show that $D\bar\Gamma_q(Y_q)$ is bounded in L^p, so $\|Y_q - \Phi_q\|_{L^p} = O(q^{-1})$. Moreover $|\mu_q - \mu_N| = O_N(q^{-1})$, and therefore

$$\|D\bar\Gamma_q(Y_q) - D\,V_N(\Phi_\infty)\|_{L^p} \le C_{N,p}\,q^{-1/2}. \tag{18.2}$$

Thus *the explicit force at the approximate forecast converges to the bare force at the terminal field.*

The error. Subtract the approximate coordinate identity $P_q = K_q Y_q + D\bar{\Gamma}_q(Y_q)$ from the exact posterior identity (12.10), $P_q = K_q \psi_q + \mathbb{E}_q[D V_N(\Phi_\infty)]$:

$$q Z_q = q C_q [D\bar{\Gamma}_q(Y_q) - \mathbb{E}_q D V_N(\Phi_\infty)]. \tag{18.3}$$

Let $G = D V_N(\Phi_\infty)$ and $G_q^0 = D V_N(\Phi_q)$, which is $\mathcal{F}_q$-measurable. Then $\|\mathbb{E}_q G - G\|_{L^p} \le \|\mathbb{E}_q (G - G_q^0)\|_{L^p} + \|G_q^0 - G\|_{L^p} \le 2\|G - G_q^0\|_{L^p} = O(q^{-1/2})$, by the cubic Lipschitz bound for $D V_N$ and the first step. With (18.2) and $\|q C_q\| \le 1$ this proves the claim. □

The proof uses no estimate uniform in N, and no concentration of the posterior: only that the observed estimator $\Phi_q = C_q P_q$, which is $\mathcal{F}_q$-measurable, converges to the hidden field. In Lecture 4 language, the forecast and the field at resolution q have the same limit at infinite resolution, and the force $F_q = K_q(\psi_q - \Phi_q)$ converges to the bare force $-D V_N(\Phi_\infty)$ (Exercise 18.1; recall $F_q = \mathbb{E}_q[-D V_N(\Phi_\infty)]$ by (12.9), so this is the conditional-expectation step of the proof).

18.2 UV stability

Now let $T \to \infty$ at fixed N in (1): the paths and the law ν_N do not depend on T, E_T and J_T increase to E_∞ and J_∞, and $\delta_{N,T,p} \to 0$ by Lemma 18.1. Only then take the supremum over N.

Theorem 18.2. (UV stability [FBSDE26]) *For q_0 large enough, depending only on γ,*

$$\sup_N (\|E_\infty^N\|_{L^p} + \|J_\infty^N\|_{L^p}) \le C_\gamma (1 + p^5), \qquad \sup_N \mathbb{E}\|Z_q^N\|_2^2 \le C_\gamma q^{-5/2}, \tag{18.4}$$

and $\sup_N \mathbb{E} \exp\{c\,(E_\infty^N + J_\infty^N)^{1/5}\} < \infty$ *for small $c > 0$.*

Proof. Monotone convergence in (1), then the representation formula (17.2) on $[q, \infty)$, whose terminal term vanishes by Lemma 18.1. The exponential moment follows from the polynomial growth by Markov's inequality with $p \asymp R^{1/5}$. □

The order of limits matters. The constant $C_{N,p}$ in Lemma 18.1 diverges with N, and we never take a supremum over N of the remainder $C_{N,p}/T$. The infinite-resolution boundary is used only qualitatively, at each lattice separately.

Exercises

Exercise 18.1. (The force at infinite resolution) Show that, at fixed N, $F_q = K_q(\psi_q - \Phi_q) \to -D V_N(\Phi_\infty)$ in every L^p, and that $F_q = K_q Z_q - D\bar{\Gamma}_q(Y_q)$. Interpret the last identity: the force observable of Lecture 4 is the explicit force at the approximate forecast, corrected by $K_q Z_q$.

Exercise 18.2. (Why the stronger boundary) Using the bound on the boundary term in the proof of Proposition 17.2, explain why $Z_T \to 0$ alone would not make it vanish, and why the rate $\|T Z_T\| = o(1)$ of Lemma 18.1 is what is needed.

19 Compactness

19.1 One Hilbert space

To compare lattices, partition $\mathbb{T}^3$ into cubic cells $\square_x$ of side N^{-1} centred at $x \in \Lambda_N$, and let

$$(\iota_N u)(z) = u(x)\ (z \in \square_x), \qquad (j_N f)(x) = N^3 \int_{\square_x} f, \qquad \mathsf{P}_N = \iota_N j_N.$$

$\iota_N : H_N \to H = L^2(\mathbb{T}^3)$ is an isometry, $j_N = \iota_N^*$, and $\mathsf{P}_N \to I$ strongly. Fields are embedded by ι_N, operators by $\iota_N \cdot j_N$. The embedded innovation has covariance P_N; in the limit it becomes a cylindrical Brownian motion on H, obtained as the limit of the lattice innovations.

19.2 Spatial suprema by a maximum principle

The energy bounds control L^4 and L^6 norms of large fields. Spatial suprema at finite resolution come for free from the elliptic structure of (15.4).

Lemma 19.1. (Supremum after closure) *Let $\varpi = \frac{3}{8}$, $\mathcal{Y}_T = \sup_{q_0 \le q \le T} q^{-\varpi} \|Y_q\|_\infty$ and $\mathcal{X}_T = \sup_{q_0 \le q \le T} q^{-\varpi} \|X_q\|_\infty$. For q_0 large,*

$$\mathcal{Y}_T \le C\,(\mathcal{X}_T + q_0^{-\varpi} \gamma^{1/2} E_T^{1/2}), \qquad \|\mathcal{X}_\infty\|_{L^p} \le C\sqrt{p}. \tag{19.1}$$

Idea. Remove the backward forcing by $\mathcal{H}_q = C_q F_q^Z$, with $\|\mathcal{H}_q\|_\infty \le C q^{-1/4} \|F_q^Z\|_2 \le C \gamma^{1/2} E_T^{1/2}$ (by Cauchy–Schwarz in q). Then $U = V - \mathcal{H}$ solves, at each frozen q, $K_q U + \frac{\gamma}{3}(U + d)^3 = f$ with $d = \mathcal{H} + X - \varrho(Y)$ and $f = \int_{q_0}^q (Y_s - X_s)\mathrm{d}s$. Constant functions of height $\max\{\|d\|_\infty, q^{-1}\|f\|_\infty\}$ are super- and subsolutions, so the lattice maximum principle bounds $\|U\|_\infty$. This gives $(1 - 2\varepsilon_q)\, y_q \le 2\,(\|\mathcal{H}_q\|_\infty + x_q) + q^{-1} \int_{q_0}^q (y_s + x_s)\mathrm{d}s$ for $y_q = \|Y_q\|_\infty$, $x_q = \|X_q\|_\infty$, where $\varepsilon_q = C\,(g_q + g_q^2)$ bounds $\|D\varrho_q\|_{\infty \to \infty}$ (Lecture 6). The weight $q^{-\varpi}$ absorbs the memory since $q^{-1-\varpi} \int_{q_0}^q s^\varpi \mathrm{d}s \le (1 + \varpi)^{-1} < 1$. The Gaussian supremum is a chaining argument: $\mathrm{Var} X_q(x) \le C q^{1/2}$ and the increments in space and scale are Hölder. □

In particular *finite-resolution* spatial suprema of Y have all moments uniformly in N. At the terminal resolution there is no such bound: Φ_∞ is a distribution.

19.3 Regularity in space and scale

On a finite scale interval $I = [q_0, Q]$ write (15.4) as the linear elliptic equation

$$K_q V_q = f_q, \qquad f_q = \int_{q_0}^q (Y_s - X_s + Z_s)\mathrm{d}s - \frac{\gamma}{3} Y_q^3. \tag{19.2}$$

The energies control $\mathbb{E} \int_I \|Y_q\|_6^6 \mathrm{d}q$ (since $|Y|^6 \le r_q^6 + |t_q|^6$) and $\mathbb{E} \int_I \|Z_q\|_2^2 \mathrm{d}q$, hence $\mathbb{E} \int_I \|f_q\|_2^2 \le C_I$, and testing with V,

$$\mathbb{E} \int_I (\|\nabla_N V_q\|_2^2 + \|V_q\|_2^2)\mathrm{d}q \le C_I.$$

So the non-Gaussian part V of the approximate forecast is in H^1, and the approximate forecast $Y = V + X - \varrho(Y)$ inherits the regularity of the free field.

Proposition 19.2. (Forward compactness) *For $0 < \sigma, \theta < \frac{1}{2}$,*

$$\sup_N \mathbb{E}\Big[\int_I \|\iota_N Y_q^N\|_{H^\sigma}^2 \mathrm{d}q + \|\iota_N Y^N\|_{H^\theta(I;H)}^2 \Big] < \infty.$$

The embedded approximate forecasts are tight in every space–time L^p, and polynomials of them converge along strongly convergent subsequences. The forward equation gives the Hölder bound $\sup_N \mathbb{E}\, \|\iota_N Y^N\|_{C^{1/4}(I;H)}^8 < \infty$, and averaging in time against the H^σ bound gives uniform Fourier tails $\lim_m \sup_N \mathbb{E} \sup_{q \in I} \|(I - \mathsf{F}_m)\, \iota_N Y_q^N\|_2^2 = 0$ (F_m the projection on frequencies $|k| \le m$); hence they are also tight in $C(I; H)$.

The spatial bound combines the H^1 bound on V, the H^σ regularity of X ($\sigma<\frac{1}{2}$), and the contraction property of ϱ; the scale regularity uses the forward equation, whose drift $b(Y)+H(Y)\,Z$ has bounded integrated square by Proposition 14.2 and the energies, and whose martingale part has bounded bracket density. The supremum bound of Lemma 19.1 upgrades L^2 compactness to every L^p.

19.4 The terminal field

The approximate forecast converges at infinite resolution in negative Sobolev spaces, uniformly in N (to the same limit as the forecast ψ). The martingale part of Y has bracket $\int H^2$, and the positive inequality $H_q^2\le\frac{C}{q}C_q$ gives

$$\int_Q^\infty \|(1-\Delta)^{-s/2}H_q\|_{\mathrm{HS}}^2\mathrm{d}q\le C\int_Q^\infty\frac{1}{q}\sum_k\frac{(1+|k|^2)^{-s}}{q+c|k|^2}\mathrm{d}q\le C_s\,Q^{-(s-\frac{1}{2})},\qquad \tfrac{1}{2}<s<\tfrac{3}{2},\tag{19.3}$$

where $q+c|k|^2$ bounds the lattice symbol from below (aliases of the cell embedding only increase the frequency). The drifts are integrable at infinity in H: the bounded-field drift is $O(q^{-5/2})$, the large-field drift is controlled by $\int q^{7/2}\|b^1\|^2\le C\gamma^3J_\infty$ and Cauchy–Schwarz, and $\int\|H_q\,Z_q\|\mathrm{d}q\le C\int q^{-1}\|Z_q\|\le C\gamma^{1/2}E_\infty^{1/2}$. More precisely, the squared drift tails beyond Q are $O(Q^{-3})$ (bounded-field), $O(Q^{-5/2})\,\gamma^3J_\infty$ (large-field) and $O(Q^{-3/2})\,\gamma E_\infty$ (the HZ term), all at most $Q^{-(s-\frac{1}{2})}$ for $s<\frac{3}{2}$. By Doob's inequality, for $\frac{1}{2}<s<\frac{3}{2}$,

$$\sup_N\mathbb{E}\sup_{t\ge Q}\|Y_t-Y_Q\|_{H^{-s}}^2\le C_s\,Q^{-(s-\frac{1}{2})}.\tag{19.4}$$

Letting $t\to\infty$ at fixed N (Lemma 18.1) this holds with Y_t replaced by Φ_∞. Together with the tightness at finite resolution, this gives:

Corollary 19.3. (Tightness of the lattice measures) *For every $s>\frac{1}{2}$ the embedded laws $\nu_N\circ\iota_N^{-1}$ are tight in $H^{-s}(\mathbb{T}^3)$, and $\sup_N\mathbb{E}\,\|\iota_N\Phi_\infty^N\|_{H^{-s}}^p<\infty$ for every p.*

For the moments: the drift integrals have all moments by (18.4), and the H^{-s} martingale has a deterministic bracket bound by (19.3) at $Q=q_0$, so the Burkholder–Davis–Gundy inequality gives all moments.

The threshold $s>\frac{1}{2}$ is that of the free field in $d=3$: the interacting field is no rougher than the Gaussian one, and at every finite resolution its non-Gaussian part V is smoother (H^1).

20 Identifying the limit

20.1 The deterministic coefficients

The coefficients M_q^N, $h_q^{[j],N}$, $H_q^N(y)$, $b_q^N(y)$ are constructed at each lattice. Their uniform bounds (Proposition 13.1) do not by themselves identify a limit: a weakly convergent vertex cannot in general be contracted against a varying covariance. The three tools are:

- *Weak vertex measures.* Derivative kernels with bounded rooted norms are signed measures with bounded mass; they converge weakly along subsequences, and the centred kernel Θ_q converges in a weak topology controlled by its first moment. The limiting coefficients solve the same fixed-point problem, whose solution is unique in its ball; so the whole sequence converges.

- *Compact sandwich.* If $G_N \to G$ in Hilbert–Schmidt norm, and B_N are symmetric, uniformly bounded and converge weakly to B, with $\sup_N \|G_N \mathsf{B}_N\| \le \theta < 1$ and $\|G\mathsf{B}\| \le \theta$, then $(I + G_N \mathsf{B}_N)^{-1} G_N \to (I + G\mathsf{B})^{-1} G$ in Hilbert–Schmidt norm. The free resolvent is the compact factor that turns weak convergence of the Hessian into strong convergence of the response.
- *Contraction against a moving covariance.* The rows of the inverses are Hölder (any exponent $a < \frac{1}{2}$); with the Hilbert–Schmidt convergence given by the sandwich, the squared kernels $(H_q^N)^2$ converge uniformly to a continuous kernel. Weakly convergent vertices of bounded variation can therefore be contracted against them, and the trace steps of the recursion pass to the limit.

With these, $H_q^N(Y_q^N) \to H_q(Y_q)$ and $b_q^N(Y_q^N) \to b_q(Y_q)$ in $L^2(I;\mathrm{HS})$ and $L^2(I;H)$ along strongly converging forecasts with the uniform supremum moments of Lemma 19.1.

20.2 The noise

The delicate point is the backward martingale. Fields converging in law do not imply that the stochastic integrals converge to stochastic integrals of the limits.

Remark 20.1. (Hidden noise) Let B be a real Brownian motion and $L_t^n = \int_0^t (1 + \sin(ns))\,\mathrm{d}B_s$. Then (L^n, B) converges in law to $(B + 2^{-1/2}\tilde{B}, B)$, with $\tilde{B}$ an independent Brownian motion: the integrands converge weakly to 1, but their squares converge weakly to $\frac{3}{2}$. A limiting FBSDE obtained this way would have an extra martingale, orthogonal to the driving noise.

To exclude this one needs *strong* compactness of the integrands. For the forward equation the integrand $H(Y)$ is an explicit function of Y, and strong convergence follows from the coefficient limits. For the backward equation the integrand is $\Xi = \mathrm{Cov} - H(Y)$, and we need strong compactness of the posterior covariance Cov_q. It comes from its exact dynamics.

Lemma 20.2. (Riccati equation of the posterior covariance) *At each N, $\mathrm{Cov}_q = D^2\mathscr{K}_q(P_q) \ge 0$ and*

$$\mathrm{d}\mathrm{Cov}_q = -\mathrm{Cov}_q^2\,\mathrm{d}q + D^3\mathscr{K}_q(P_q)[\mathrm{d}W_q]. \tag{20.1}$$

Proof. Differentiate the equation of Proposition 12.3(c) twice in p: $(\partial_q + \frac{1}{2}\Delta_p + D\mathscr{K}\cdot D_p)\,D^2\mathscr{K} = -(D^2\mathscr{K})^2$, and apply Itô's formula along $\mathrm{d}P = D\mathscr{K}(P)\mathrm{d}q + \mathrm{d}W$. □

This is (8.9) for $\partial_q R = 1$: conditional variance is resolved at the rate Cov^2. For a smooth test f, the process $\langle f, \mathrm{Cov}_q f\rangle \ge 0$ is a nonnegative supermartingale, with drift $-\|\mathrm{Cov}_q f\|^2$, and initial value $\mathrm{Var}_{\nu_N}\langle \Phi_\infty, f\rangle$, bounded uniformly in N by Corollary 19.3. A nonnegative supermartingale has controlled oscillations: after stopping at the first time it exceeds a level K (an event of probability at most C_f/K), its decreasing part A and the bracket of its martingale part J satisfy $\mathbb{E}\,A_Q \le C_f$ and $\mathbb{E}[J]_Q \le K^2 + 2KC_f$, which gives time regularity in $W^{a,1}$, $a < \frac{1}{2}$, and hence strong compactness of each finite block $\langle f_i, \mathrm{Cov} f_j\rangle$. Spatial tails come from the isometry (recall $\psi_{q_0} = \mathbb{E}\,\Phi_\infty = 0$, ν_N being even), with symmetry exchanging input and output tails, $\mathbb{E}\int_I \|(I - \mathsf{F}_m)\mathrm{Cov}_q\|_{\mathrm{HS}}^2 \mathrm{d}q = \mathbb{E}\|(I - \mathsf{F}_m)\,\psi_Q\|_2^2$, and from the terminal averaging $\psi_Q = \mathbb{E}_Q \int_L^{L+1} (Y_s + Z_s)\mathrm{d}s$, using the H^σ bound of Proposition 19.2 and $\mathbb{E}\|Z_s\|_2^2 \le C\,s^{-5/2}$ (first $m \to \infty$, then $L \to \infty$). Altogether Cov^N is tight in strong $L^2(I;\mathrm{HS})$. Since $S \mapsto \int S\,S^*$ is continuous from $L^2(I;\mathrm{HS})$ to $C(I;\text{trace class})$, the brackets of ψ^N are tight in $C(I)$; a continuous martingale with tight brackets and uniformly small maximal Fourier tails (from the terminal averaging above and Doob's inequality) is tight in $C(I;H)$. Hence ψ^N, and $Z^N = \psi^N - Y^N$, are tight in $C(I;H)$. Strong convergence of the integrands gives convergence of the brackets $\int \mathrm{Cov}\mathrm{Cov}^*$ and of the covariations with W, and then:

Proposition 20.3. (Identification of the martingales [FBSDE26]) *Along a subsequence, on a common filtered probability space, the limit of the forecast satisfies $\psi_q = \int_{q_0}^{q} \mathrm{Cov}_s \, \mathrm{d}W_s$ with W a cylindrical Brownian motion for the joint filtration, and Y, Z, X satisfy*

$$Y_q = \int_{q_0}^{q} [b_s(Y_s) + H_s(Y_s)\, Z_s]\mathrm{d}s + \int_{q_0}^{q} H_s(Y_s)\, \mathrm{d}W_s,$$
$$Z_q = -\int_{q_0}^{q} [b_s(Y_s) + H_s(Y_s)\, Z_s]\mathrm{d}s + \int_{q_0}^{q} \Xi_s \, \mathrm{d}W_s, \qquad X_q = \int_{q_0}^{q} C_s \, \mathrm{d}W_s. \tag{20.2}$$

The argument is: the limiting ψ is a continuous square-integrable martingale for the joint filtration (fixed-time martingale identities pass to the limit with uniform integrability); its bracket and its covariation with W are those of $\int \mathrm{Cov}\, \mathrm{d}W$; so $\psi - \int \mathrm{Cov}\, \mathrm{d}W$ has zero bracket and vanishes. No extra martingale survives. The same argument for the forward equation identifies its noise with the same W, and identifies the limiting Gaussian part X as $\int C \, \mathrm{d}W$, not a new Gaussian field.

20.3 The result

Theorem 20.4. (Subsequential construction of Φ^4_3 [FBSDE26]) *Let $\lambda > 0$ and q_0 large enough. Along a subsequence of lattices:*

a) the embedded processes (Y^N, Z^N, ψ^N, X^N) converge in law in $C(I;H)$ for every finite I, together with the embedded innovations W^N in $C(I;H^{-r})$, $r > \frac{3}{2}$, and the integrands (Cov^N, Ξ^N) in strong $L^2(I;\mathrm{HS})$, jointly and with an almost sure representation;

b) the limit solves the FBSDE (20.2) on $[q_0, \infty)$, with $Y_{q_0} = Z_{q_0} = 0$, the deterministic continuum coefficients b, H, limits of those of Lecture 5 and unique in their ball, an H-cylindrical Brownian motion W for the joint filtration, obtained as the limit of the lattice innovations, a symmetric nonnegative integrand $\mathrm{Cov} = H(Y) + \Xi$, and $\mathbb{E}\|Z_q\|_2^2 \le C q^{-5/2}$;

c) there is a terminal field Φ with $Y_q \to \Phi$ in H^{-s}, almost surely and in mean square, for every $s > \frac{1}{2}$. Its law ν is the weak limit of the lattice measures ν_N along the subsequence, and for smooth h,

$$\langle Y_q + Z_q, h\rangle = \mathbb{E}[\langle \Phi, h\rangle \mid \mathcal{F}_q]: \tag{20.3}$$

the forecast remains the conditional mean of the terminal field;

d) the limiting thresholded energies E_∞, J_∞ (built with $X = \int C \, \mathrm{d}W$) satisfy the bounds of Theorem 18.2 (Fatou).

So the posterior structure survives the continuum limit: $\psi = Y + Z$ is a closed observable, the conditional mean of the terminal field; Y is computed from the explicit action; and the error Z is small at high resolution. This is the FBSDE of Lecture 4, in forecast coordinates, for a subsequential limit ν of the renormalised lattice measures, with the full noise retained.

20.4 The renormalised field equation

The terminal law satisfies an integration by parts formula in which the renormalised cube appears only tested in expectation.

Theorem 20.5. (Integration by parts) *Let $h, h_1, \dots, h_k$ be smooth and $G(\varphi) = g(\langle \varphi, h_1\rangle, \dots, \langle \varphi, h_k\rangle)$ with $g \in C_b^\infty(\mathbb{R}^k)$, and let $D_h G(\varphi) = \frac{\mathrm{d}}{\mathrm{d}a} G(\varphi + a\,h)|_{a=0}$. Along the subsequence of Theorem 20.4 (recall $C = (m^2 - \Delta_N)^{-1}$ with $m^2 = q_0$),*

$$\lim_N \mathbb{E}\left[G(\iota_N \Phi^N_\infty)\, \langle \mu_N \Phi^N_\infty + \lambda (\Phi^N_\infty)^3, j_N h\rangle_{H_N}\right] = \mathbb{E}_\nu\left[D_h G(\Phi) - G(\Phi)\langle \Phi, (q_0 - \Delta)\, h\rangle\right]. \tag{20.4}$$

Proof. At fixed N, Gaussian integration by parts in the density $\mathrm{e}^{-V_N-\frac{1}{2}\langle\varphi,K_{q_0}\varphi\rangle}$ gives, with $x=\Phi^N_\infty$ and $h_N=j_Nh$,

$$\mathbb{E}\,[G(\iota_N x)\langle\mu_N x+\lambda x^3,h_N\rangle]=\mathbb{E}\,[D_{h_N}(G\circ\iota_N)(x)]-\mathbb{E}\,[G(\iota_N x)\langle x,K_{q_0}h_N\rangle].$$

Cell averaging commutes with lattice translations, so $K_{q_0}j_Nh=j_N(q_0-\Delta^{\mathrm{tr}}_N)h$ with Δ^{tr}_N the finite-difference Laplacian acting on smooth functions, which converges to Δh in every Sobolev norm. The right side converges by Corollary 19.3 (weak convergence in H^{-s} and uniform moments), and the derivative term is a finite sum with converging coefficients $\langle j_Nh,j_Nh_i\rangle\to\langle h,h_i\rangle$. □

This is the terminal case of the Dyson–Schwinger equation of Theorem 11.1, (11.3), for the limiting theory. It does not say that $\mu_N\Phi+\lambda\Phi^3$ converges as a random distribution: only its expectation against cylinder functions is identified. This is consistent with the expectation of Lecture 4 (Figure 7.1 and the remark following it) that the limiting force is an observable without a terminal representative; it is not proved here that no such representative exists. At every *finite* resolution, on the other hand, the force is explicit: $F_q=K_qZ_q-D\bar\Gamma_q(Y_q)$ (Exercises 18.1 and 20.1).

20.5 What is not proved, and where to go next

- *Uniqueness.* The limit is subsequential. Uniqueness of the limiting FBSDE, or of ν, would give convergence of the whole sequence and independence of the regularisation.
- *Small mass* (*strong coupling*). The construction needs $\gamma q_0^{-1/2}$ small. By scaling, it is equivalent to the weak-coupling regime $\lambda/m\ll1$ (unit mass on a torus of side m). Removing it requires a priori estimates that do not rely on the smallness of the feedback coefficient γ^2/q_0, and a coefficient construction outside the perturbative ball ($g_{q_0}=\gamma q_0^{-1/2}$ small in Proposition 13.1).
- *Axioms.* Reflection positivity holds at every lattice, for reflections in lattice and mid-lattice planes (nearest-neighbour action), and passes to ν for every reflection in a coordinate plane, by approximating the plane with lattice planes (standard; Exercise 20.4). Rotation invariance (absent on the torus) and clustering need infinite volume.
- *Infinite volume.* The force, its Burgers flow, and differences of free energies are intensive (Section 9.2). The estimates of Lecture 6 must then be made local, with weights in space and constants per unit volume. A new difficulty appears in the backward equation: its integrand Ξ is a nonlocal operator, and the backward error at a point can receive noise from arbitrarily far away. Uniform-in-volume coefficient bounds and local energy moments are themselves open; they could give an infinite-volume Gibbs subsequence. Controlling the "noise from infinity" (together with local compactness of the Hessian blocks) is the main additional step needed to retain the full FBSDE without an orthogonal martingale [FBSDE26].

Summary of the course. The seven lectures went from the Wightman axioms to the Euclidean formulation (Lecture 1), to the free field (Lecture 2), to stochastic quantisation and the scale decomposition (Lecture 3), and to the three faces of an interacting field: the Polchinski equation, the FBSDE and the variational problem (Lecture 4). In Lectures 5–7 the FBSDE in forecast coordinates, together with an explicit solution of the Wetterich equation up to an integrable defect, gave a construction of Φ^4_3 in which every object has a probabilistic meaning: the forecast is a conditional mean, the response is a conditional covariance, the Brownian motion is an innovation, and UV stability is the statement that an explicit action predicts the conditional mean up to an error that is small at high resolution.

Lecture 4 (formal)	Lectures 5–7	status in the limit
forecast $\psi_t = \mathbb{E}_t \Phi_T$	$\psi = Y + Z$	closed martingale, (20.3)
response $H_t = \mathrm{Cov}_t$	$\mathrm{Cov} = H(Y) + \Xi$	strong $L^2(\mathrm{HS})$, Riccati
Γ_t, Wetterich	$\bar{\Gamma}_q$ explicit, defect e	deterministic coefficients
force F_t	$K_q Z_q - D\bar{\Gamma}_q(Y_q)$	observable at finite q
DS equation	Theorem 20.5	tested in expectation

Exercises

Exercise 20.1. (**The force observable**) Show that $\langle F_q^N, j_N h\rangle = \langle \psi_q^N, K_q j_N h\rangle - \langle W_q^N, j_N h\rangle - \int_{q_0}^q \langle \psi_s^N, j_N h\rangle \mathrm{d}s$. Deduce that, along the subsequence of Theorem 20.4, these converge, with uniformly bounded moments, to $\langle F_q, h\rangle = \int_{q_0}^q \langle (\mathrm{Cov}_s K_s - I)\, h, \mathrm{d}W_s\rangle$, a martingale for the limiting filtration (note $F_{q_0} = 0$, since ν_N is even). (Cross-check: $F_q = K_q Z_q - D\bar{\Gamma}_q(Y_q)$, with Lemma 19.1 and Theorem 18.2, gives a uniform second-moment bound.) Why does this not give a terminal value F_∞?

Exercise 20.2. (**Hidden noise**) Prove the claim of Remark 20.1. Where would the argument of Proposition 20.3 fail if only weak convergence of Cov^N were known?

Exercise 20.3. (**Riccati and positivity**) For a Gaussian $\nu_N = \mu^C$, solve (20.1) and check that $\mathrm{Cov}_q = C_q$. For a general ν_N, show that TrCov_q is a supermartingale.

Exercise 20.4. (**Reflection positivity**) Show that ν_N is reflection positive for reflections in lattice and mid-lattice planes (nearest-neighbour Gaussian times a local potential). Deduce reflection positivity of ν for every coordinate plane $\{x_1 = a\}$: take cylinder functions of tests supported in $\{a < x_1 < a + \frac{1}{2}\}$ (one of the two components of the torus minus the fixed planes $\{x_1 = a\}$ and $\{x_1 = a + \frac{1}{2}\}$ of θ_a), cell-average them, and use lattice planes $a_N \to a$, so that for large N the averaged tests lie on the correct side. Since lattice and mid-lattice reflections map cells to cells, $\theta_{a_N} j_N f = j_N \theta_{a_N} f$, so the reflected tested variables are $\langle \iota_N \Phi, \theta_{a_N} f\rangle$, with $\theta_{a_N} f \to \theta_a f$ in every H^s.

Exercise 20.5. (**Negative Sobolev threshold**) Verify (19.3), and show that at $s = \frac{3}{2}$ the bound becomes $Q^{-1} \log(2 + Q)$.

References

[**FBSDE26**] M. Gubinelli. The massive Φ_3^4 field through forward–backward equations. Course notes, October 2026.